

Lab 4: ROM-based Game Access Control on FPGA

User Manual

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1. Description

A game that can be played by two players on the DE0-CV FPGA board. Each player has one set of four slide switches on the board corresponding to a 4-bit binary number. Player 1 switches are the four rightmost switches (SW3-SW0) which SW0 is the least significant bit. Player 2 switches are the four leftmost switches (SW9-SW6) which SW6 is the least significant bit. Player 1 and Player 2's numbers are displayed on the rightmost 7-segment display and the second to leftmost display in hexadecimal value respectively. The sum of the two players' numbers is displayed on the third 7-segment display from the right in hexadecimal value. The leftmost LED is lit if the sum of the two numbers equal 4'b1111 or its right-adjacent LED is lit if otherwise.

Players have their own LOAD button that will display their respective number on the switches on the 7-segment displays. Before any player can load their number, the system password must be entered using Player 2's set of switches. If the password is successfully entered, the rightmost LED is lit up indicating logged-in status or the left-adjacent LED is lit if a correct password has not been entered. Once logged in and the password LOAD button is pressed again, players will be logged out. They cannot load their numbers until logged in again. The RESET button will log players out and set the board back to default settings.

The rule of the game is that Player 1 sets their number using the switches then Player 2 needs to set a number (one-time only) that the sum of both numbers equals 4'b1111 without looking at Player 1's switches. If Player 2 gets the right number, they win the round. If Player 2 gets the wrong number, Player 1 wins the round. The players switch roles after every round as Player 2 now sets a number for Player 1 to match.

In Lab 3, Player 2 is replaced by a hardware-based random number generator (RNG). Instead of manually selecting a number, the system generates a 4-bit random value (0–15) using a counter driven by the 50 MHz clock. The player presses and holds a dedicated button to start the counter and releases it to capture the random value, which is then displayed for the player to match. This button is not debounced so that the length of the press determines the generated number.

The password protection and LOAD functionality remain the same as in the previous design. Before the correct password is entered, pressing the random number generation button or LOAD button will not produce any output. Once logged in, the player can generate random numbers and load their inputs normally.

A game timer is also introduced in Lab 3. The system includes a countdown timer starting at 99 seconds, displayed in base-10 on the 7-segment displays. Before logging in, the timer shows "00." After entering the correct password, the timer is set to 99 seconds and begins counting down when the GAME button is pressed. The player must complete as many correct matches as possible before the timer reaches zero.

When the timer expires, the game ends and no further inputs or random number generations are allowed. Pressing the GAME button resets the timer back to 99 seconds, and pressing it again starts a new game session. The logout function is disabled in this version since the same button is used to control the game timer.

Lab 4 introduces two additional implementation features while keeping the same user interface and interaction sequence. First, the timer control function is separated from the password authentication logic, and the original millisecond-based timer is replaced with a more cost-efficient design based on a Linear Feedback Shift Register (LFSR). Second, player

authentication is modified so that the default password is stored in on-chip ROM and fetched during verification rather than being hardcoded in logic. The password entry sequence and game start/restart behavior remain consistent with Lab 3. As optional bonus features, the system supports logging out by pressing the Player LOAD button when the game is not in an active timed session, and after logout all gameplay functions (including RNG and timer activity) are disabled until the user logs in again. Additionally, the RNG's internal 4-bit counter can be replaced with a more cost-efficient 4-bit LFSR-based counter while maintaining the same press-and-hold capture behavior.

Binary (switches)	Hexadecimal (display)	Decimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

Table 1: Hexadecimal (on display) and binary (switches) conversion to decimal

2. FPGA Board Overview

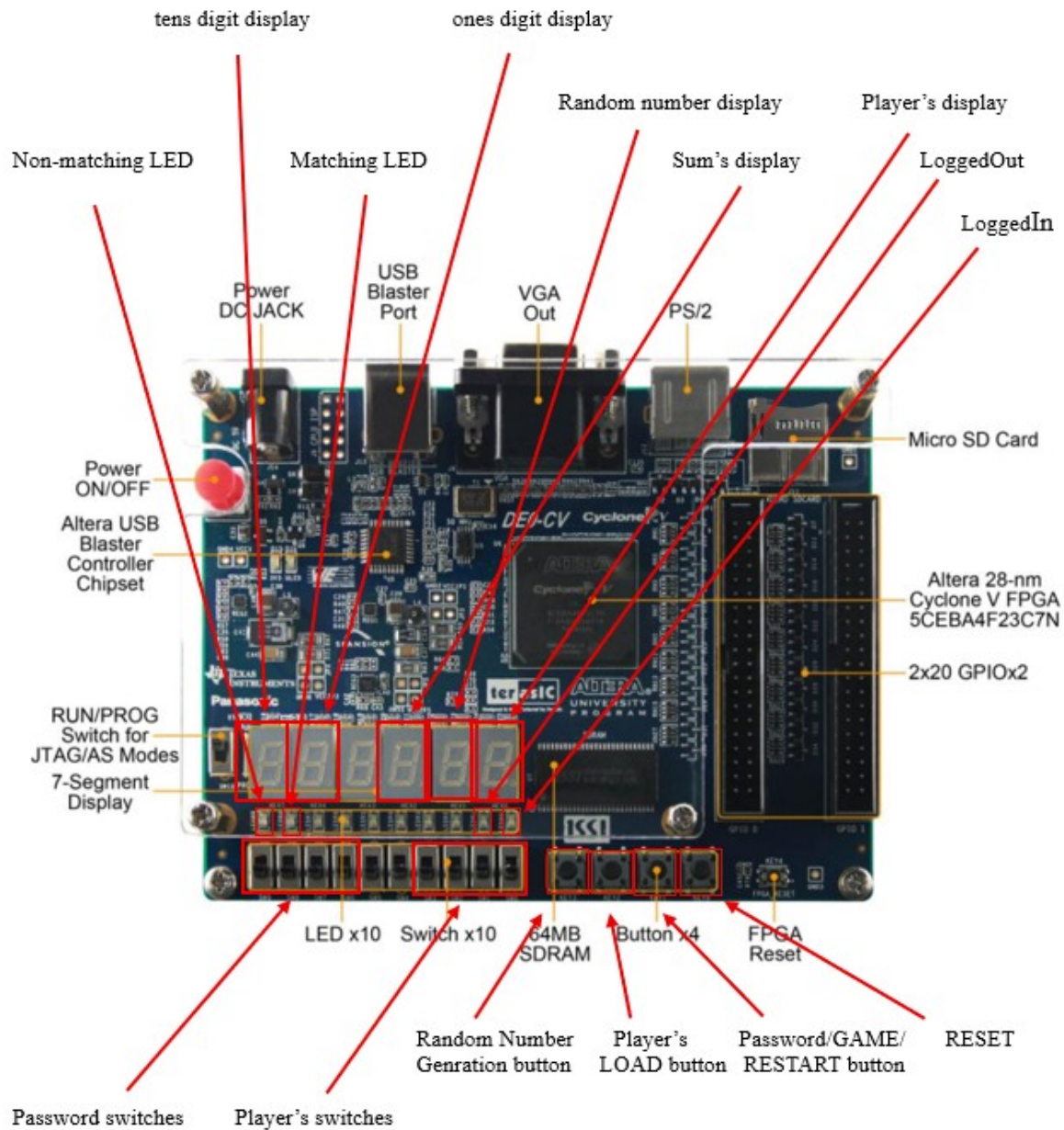


Figure 1: DE0-CV FPGA board with annotated Player' switches, displays, and LEDs

3. Step-by-step Game Rule

3.1. Authentication

Step 1: FPGA is powered and in default settings showing 0 on all three displays. The leftmost LED is lit depicting non-matching condition while the next-to-rightmost LED is lit indicating logged out status. The password is 1032.

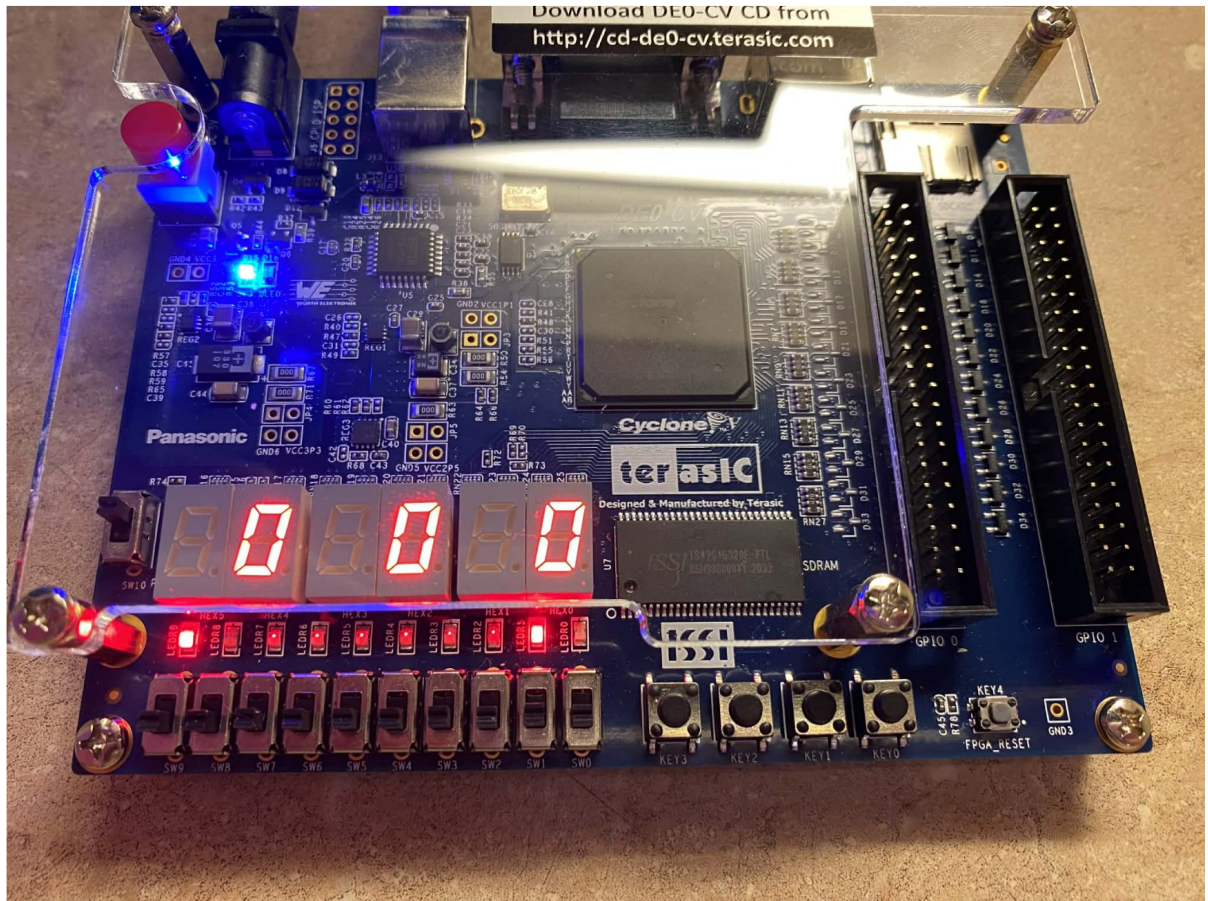


Figure 2: FPGA default settings

Step 2: The first password digit (1'd1 or 4'b0001) is entered using Player 2's switches. The same two LEDs are still lit.

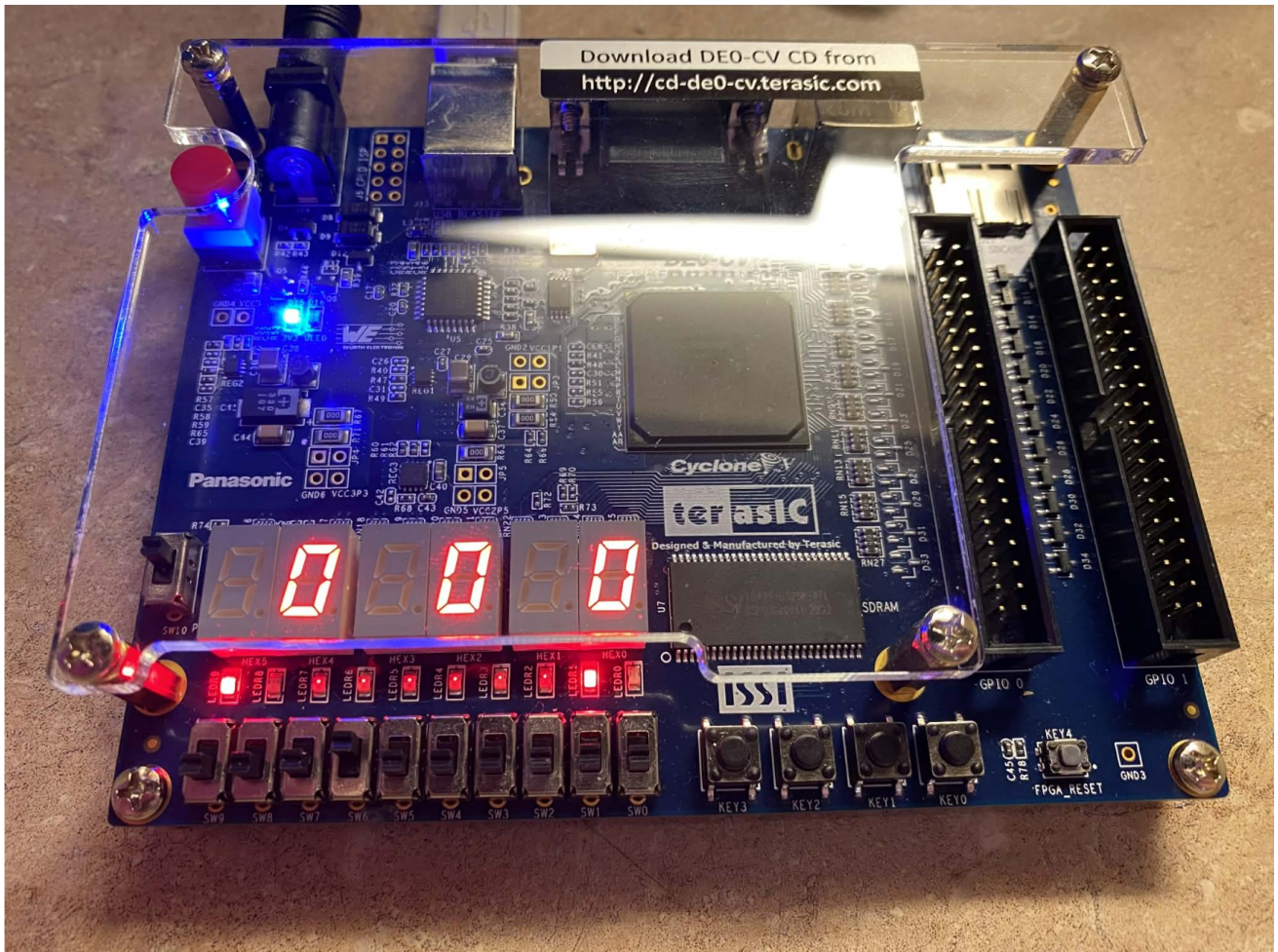


Figure 3: First password digit entered

Step 3: The second password digit (1'd0 or 4'b0000) is entered using Player 2's switches. The same two LEDs are still lit.

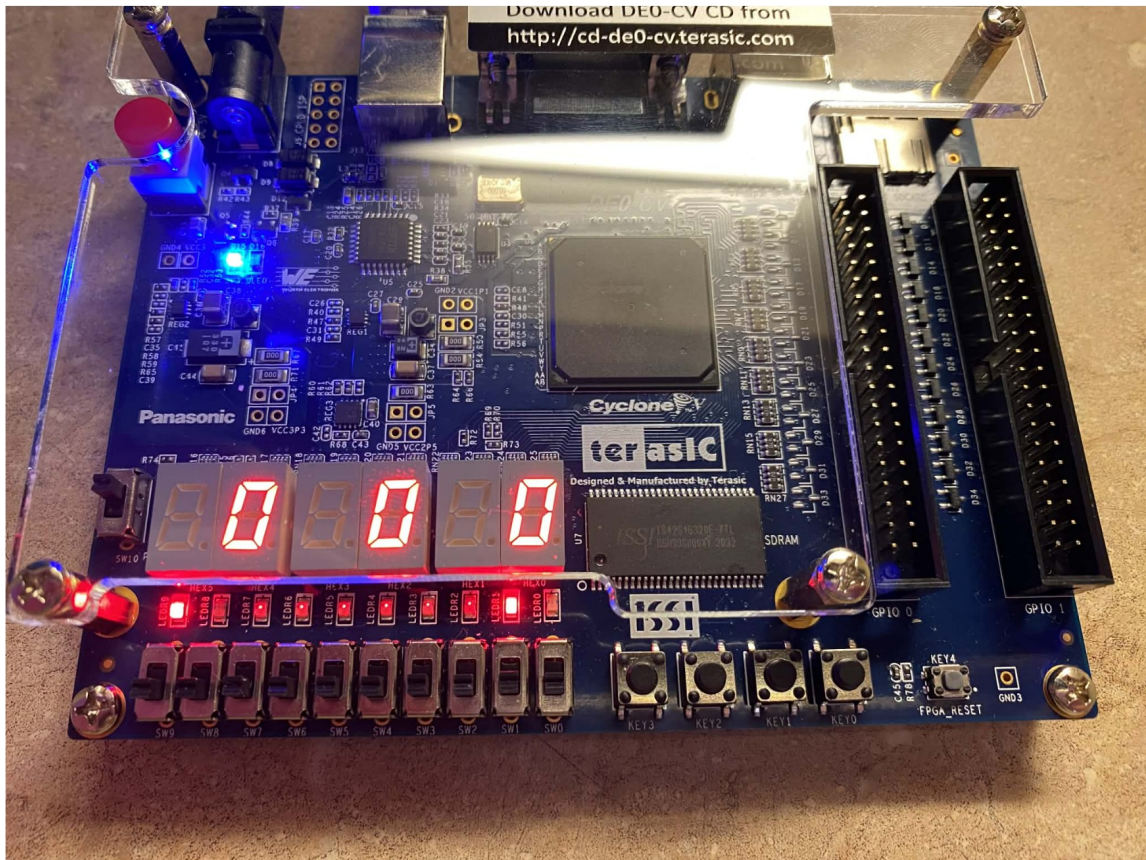


Figure 4: Second password digit entered

Step 4: The third password digit (1'd3 or 4'b0011) is entered using Player 2's switches. The same two LEDs are still lit.

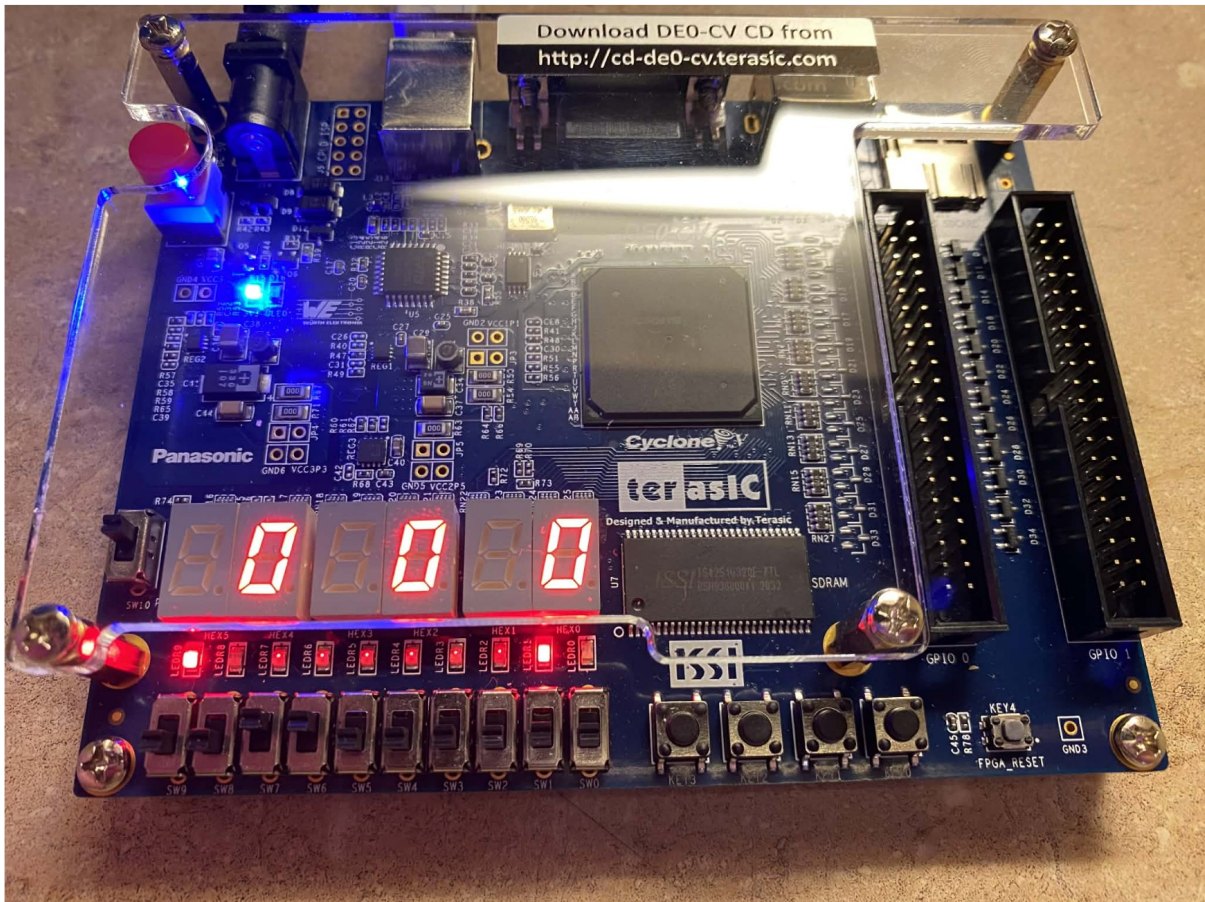


Figure 5: Third password digit entered

Step 5: The last password digit (1'd2 or 4'b0010) is entered using Player 2's switches. The same two LEDs are still lit.

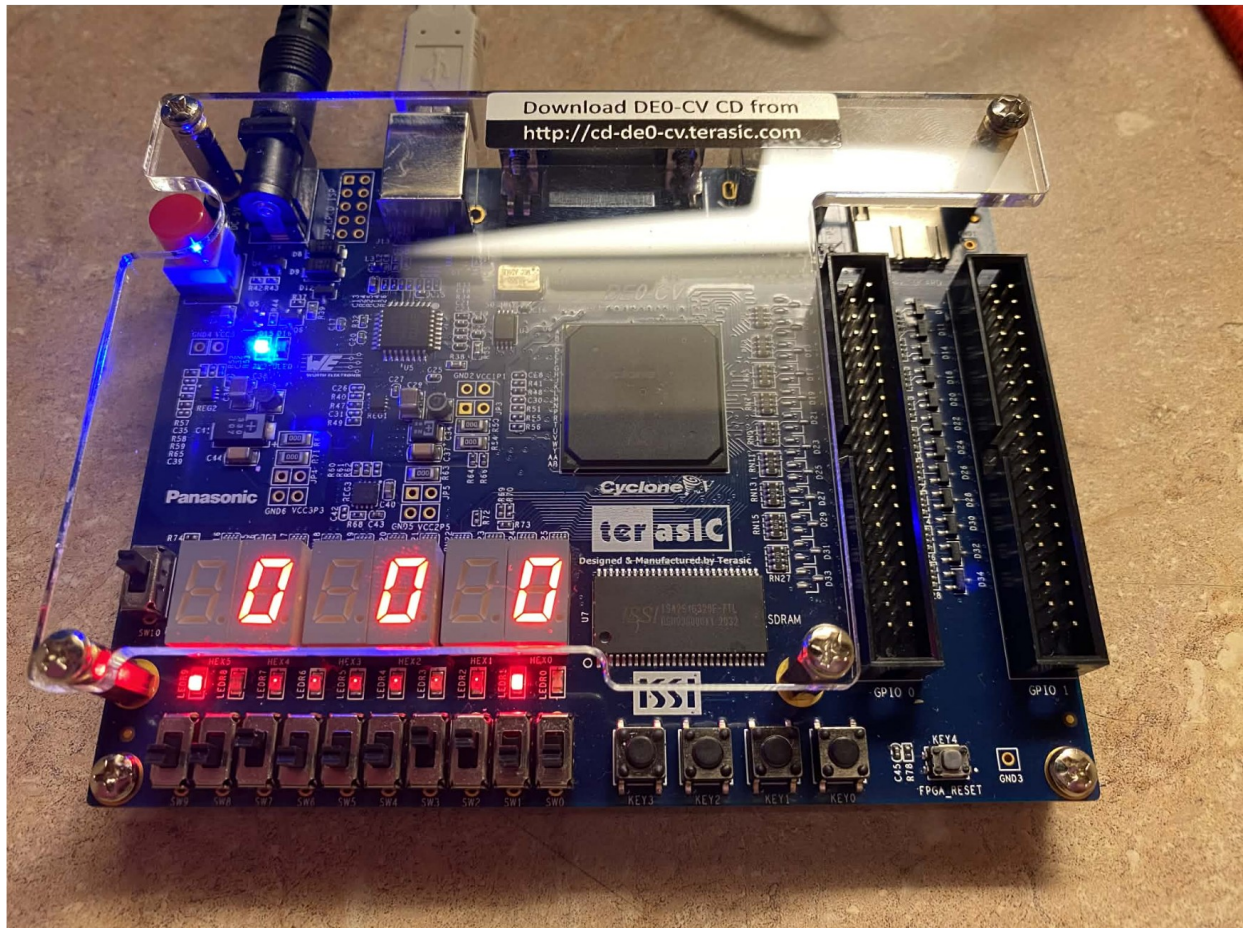


Figure 6: Last password digit entered

Step 6: Password authentication is still the same and after successfully entering all four digits. The system is logged in as showing:

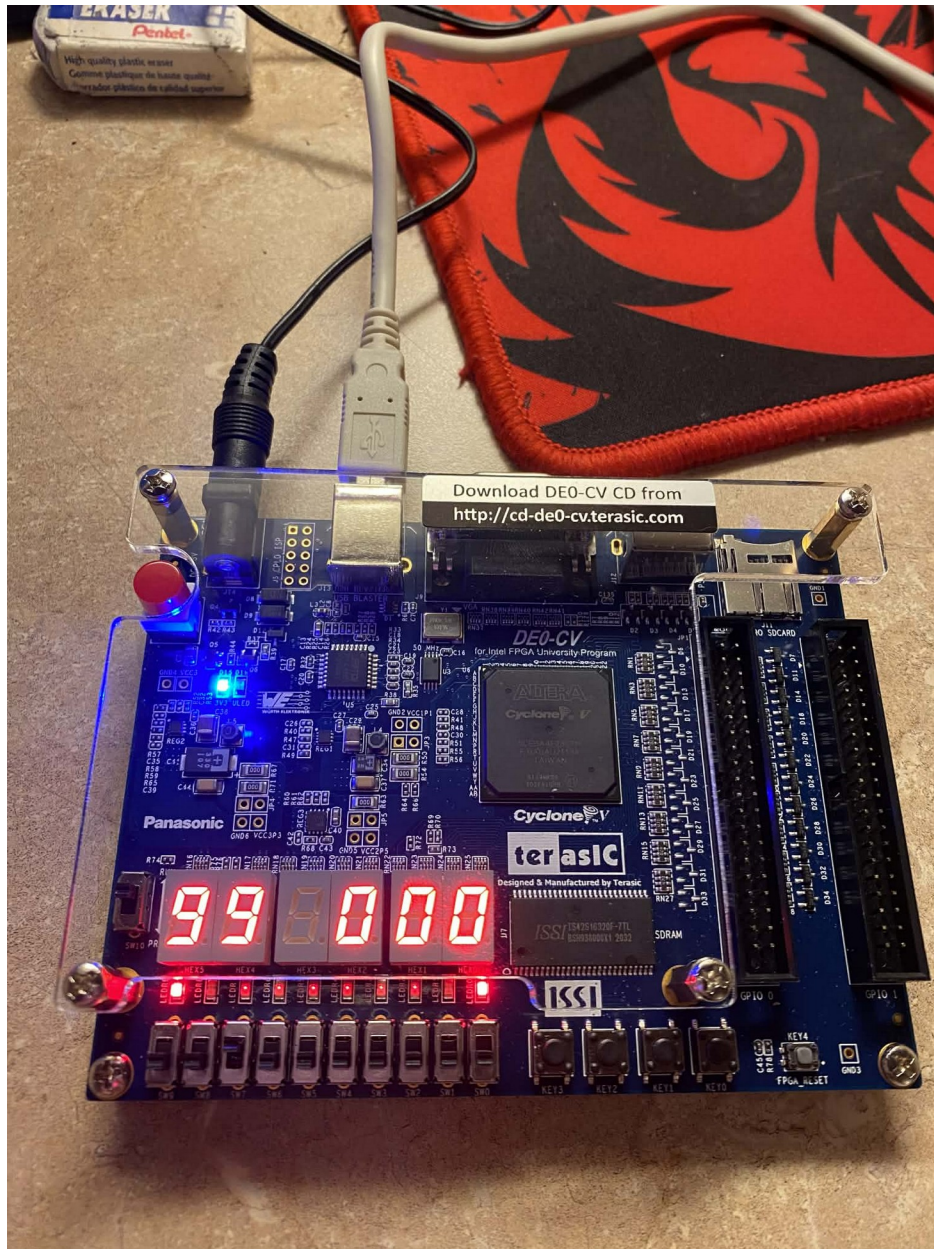


Figure 7: Logged in status

Step 7: Pressing the Game button will have the timer start counting down.

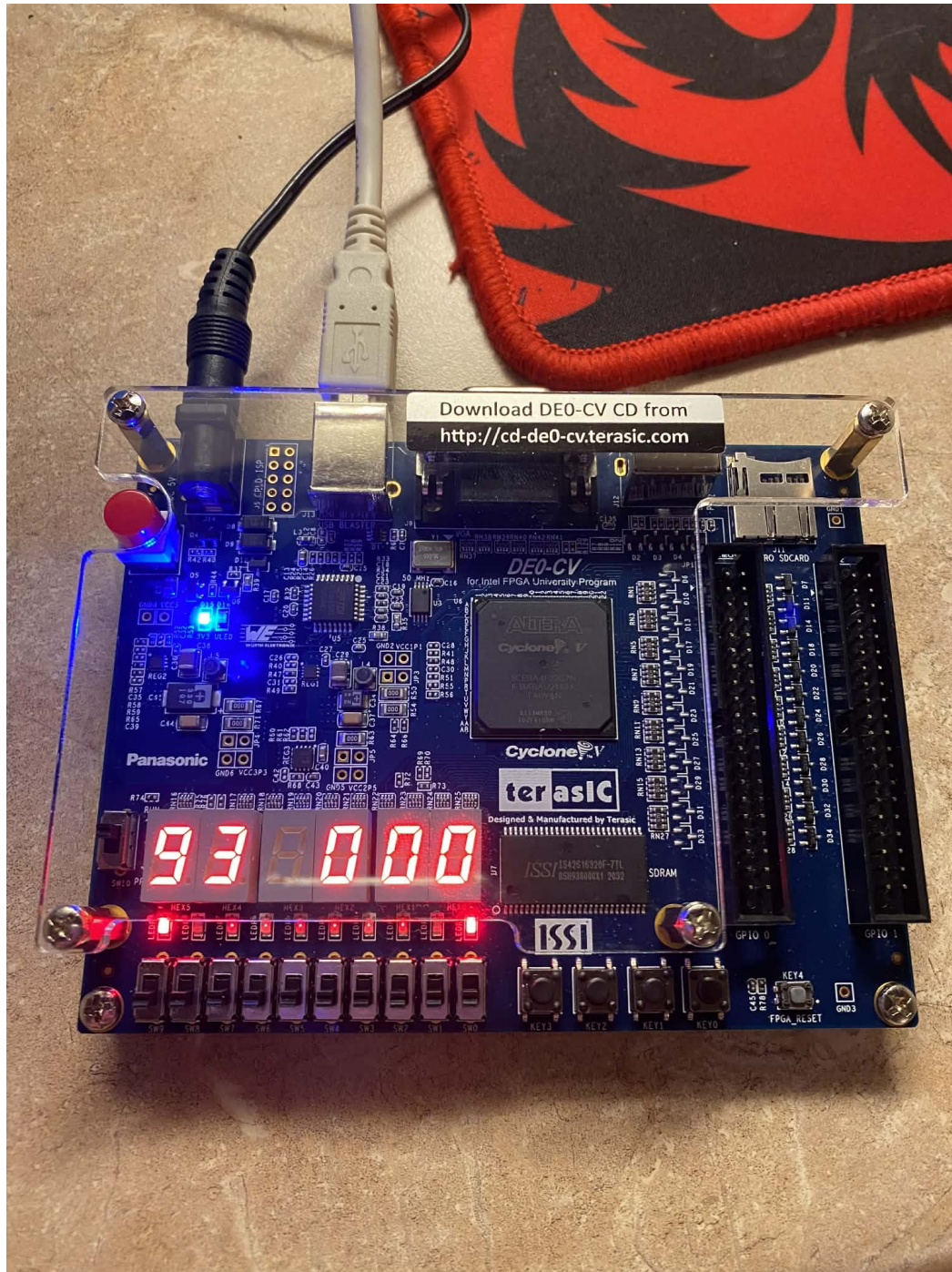


Figure 8: Timer starts counting

Step 8: Pressing the random number generation button will display a random number on the next-to-rightmost display for Player to match.

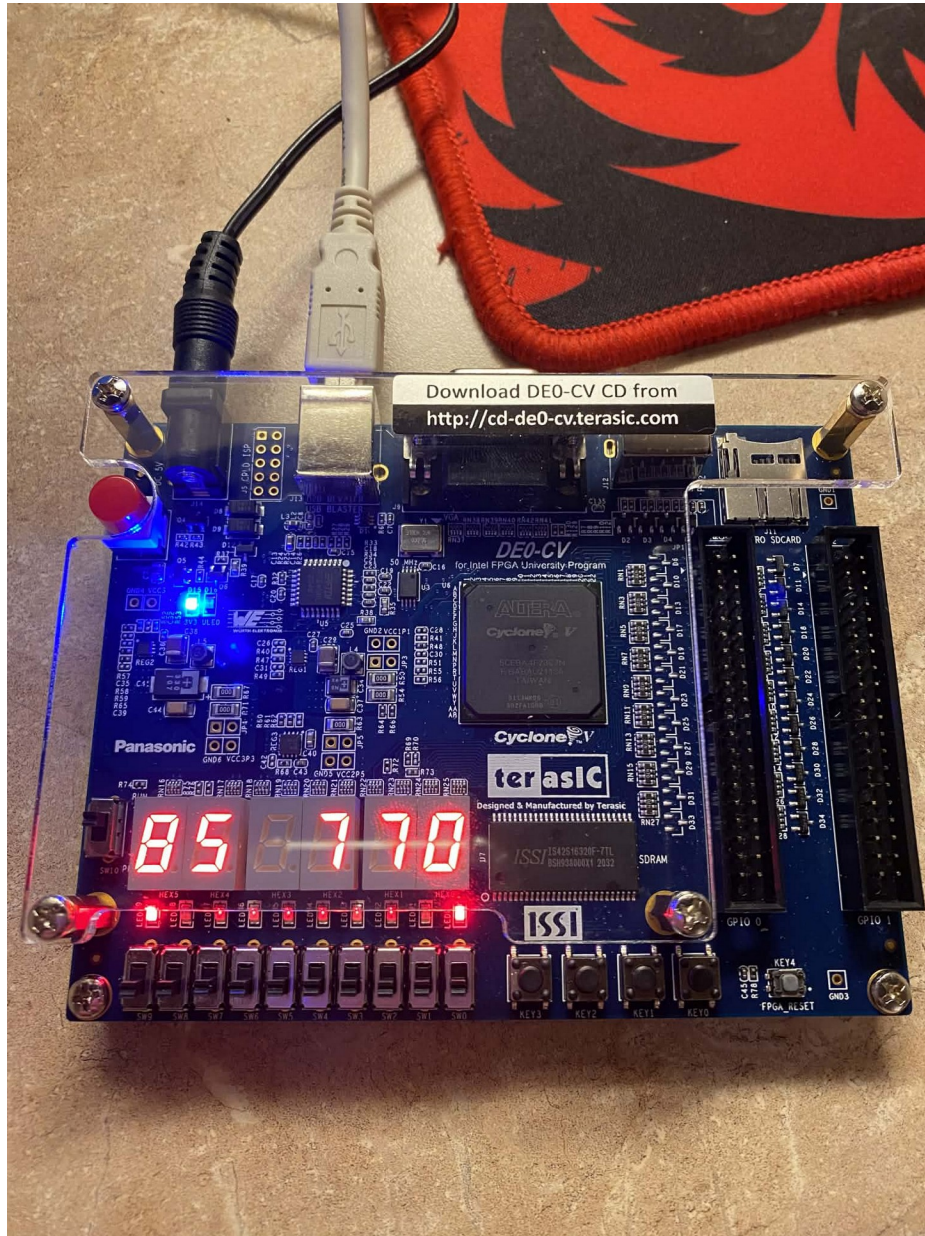


Figure 9: Random number is displayed

Step 9: Player enters a matching number (8) and pushes the LOAD button to load their number. The third-from-rightmost display shows the sum of two numbers (F means a matching pair is found). Next-to-leftmost LED is lit indicating matching condition.

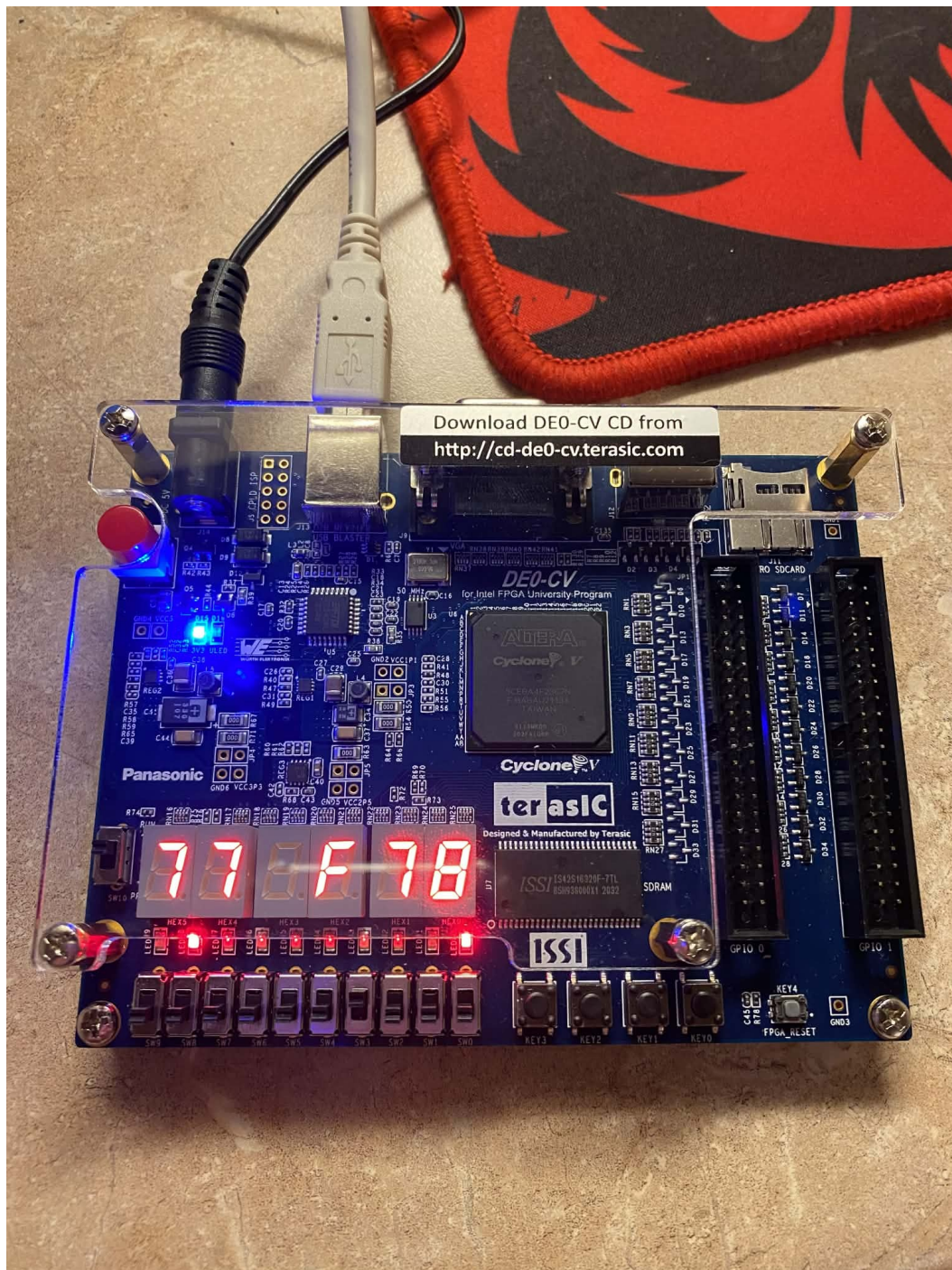


Figure 10: Matching number is found and LED is lit

Another matching pair is found. Timer still ticking down.

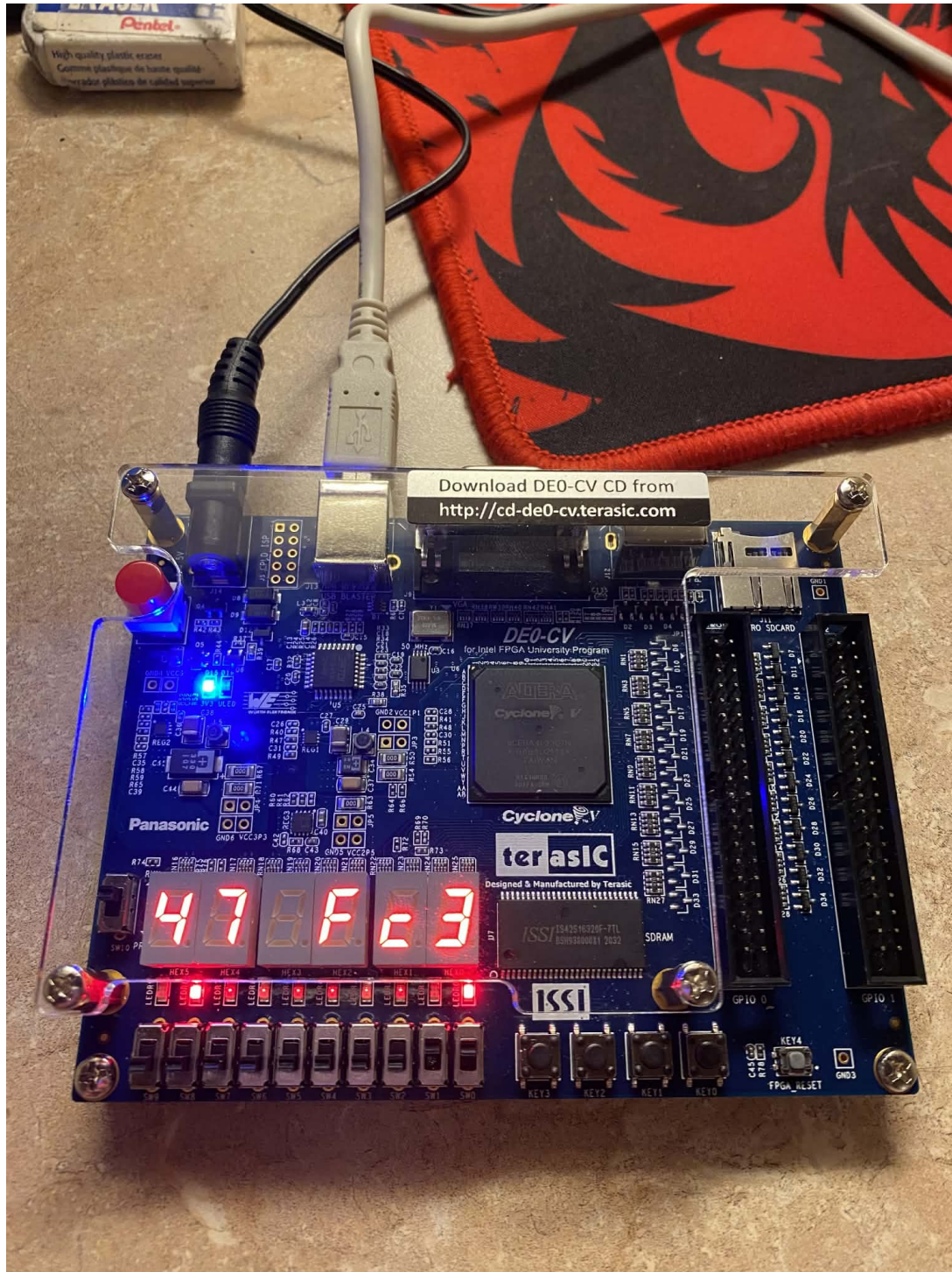


Figure 11: Another matching pair is found

Step 10: When timer hits 00, gameplay is disabled.

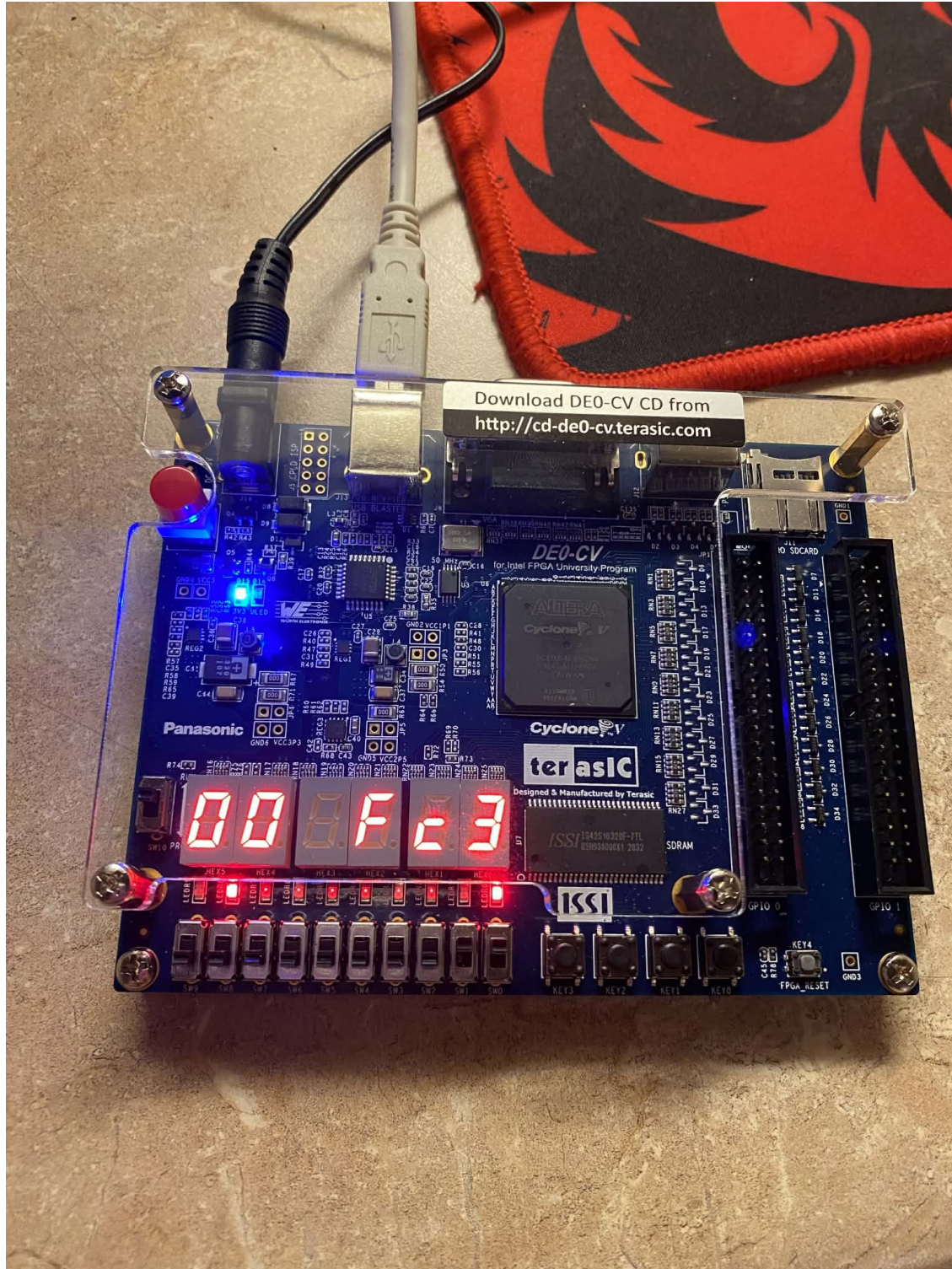


Figure 12: Game over status

Step 11: Pressing the Game/GameRestart button again will restart the timer. Pressing that same button again will start the timer and Player can start playing again.

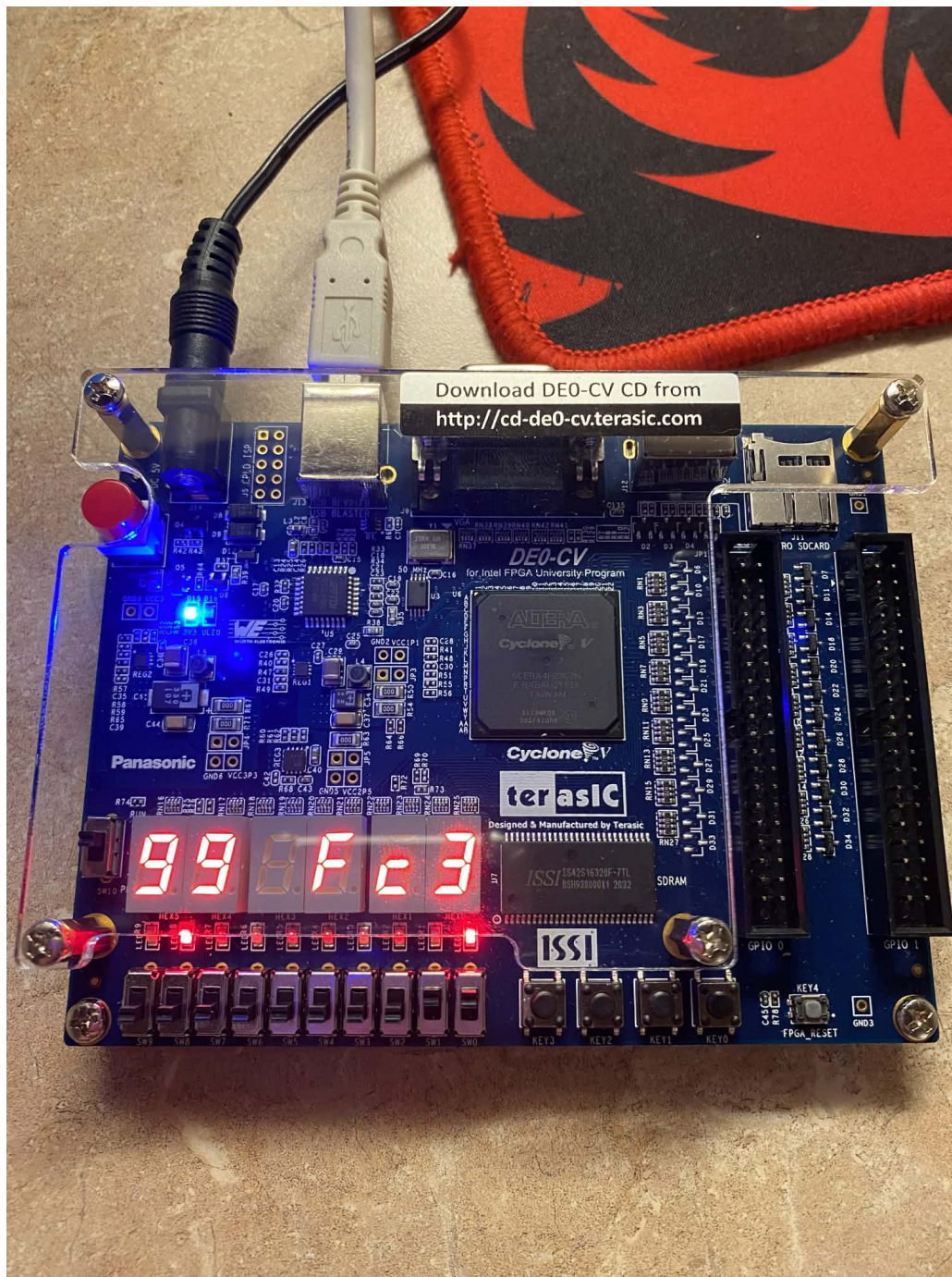


Figure 13: Timer is reloaded and game is restarted

Step 12 (log out function): Pressing the Player LOAD button outside of gameplay state will trigger a log out request.

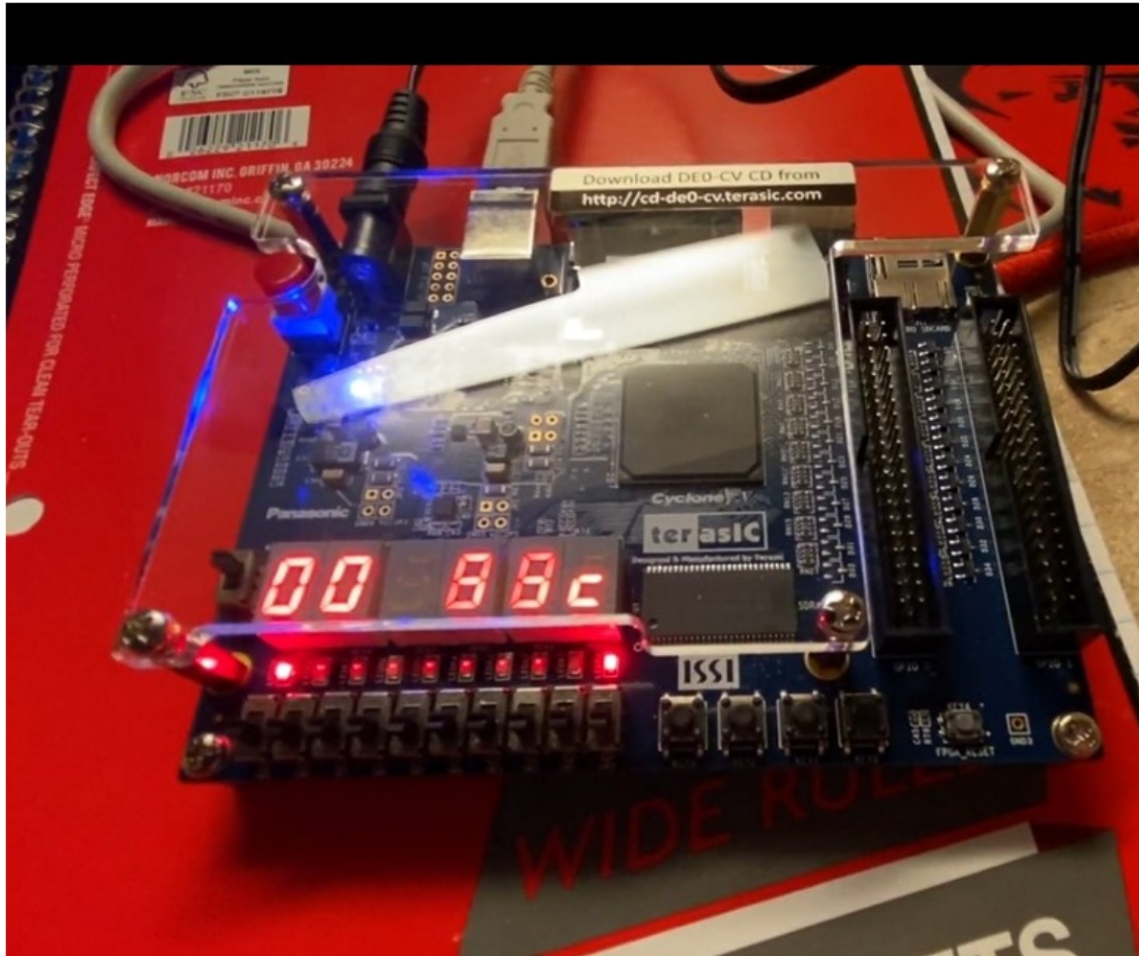


Figure 14: Current status is logged in after a round was over

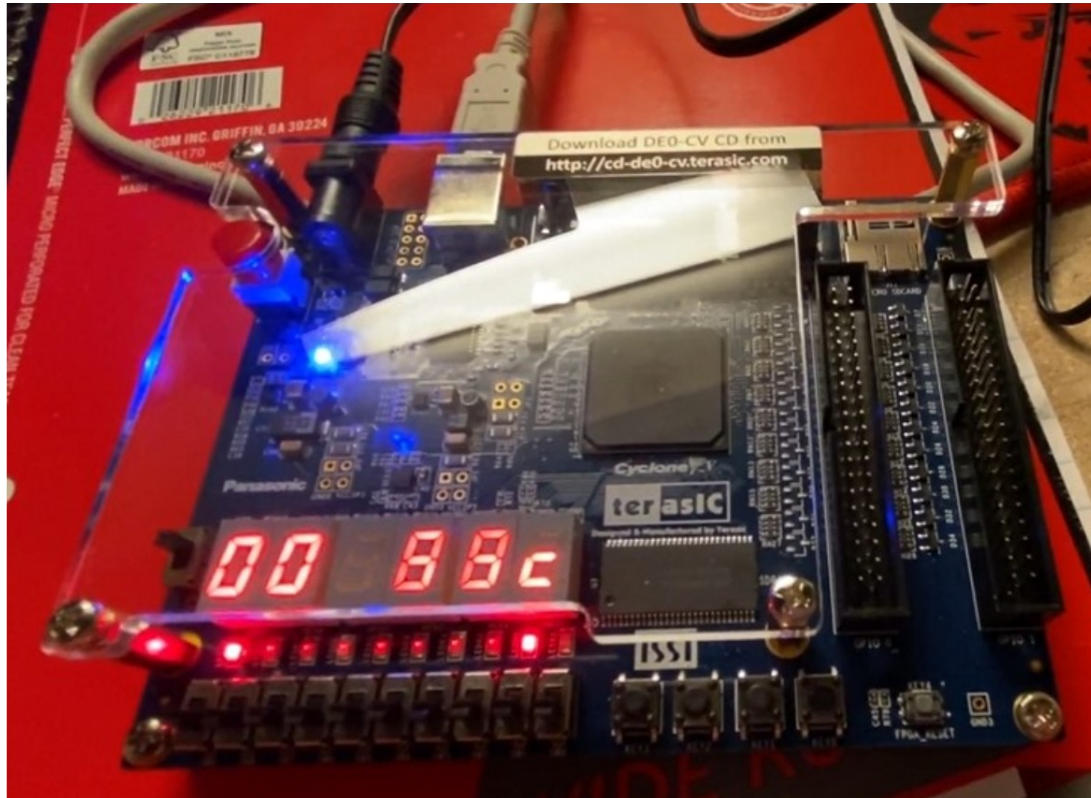


Figure 15: Current status is logged out after Player LOAD button was pressed

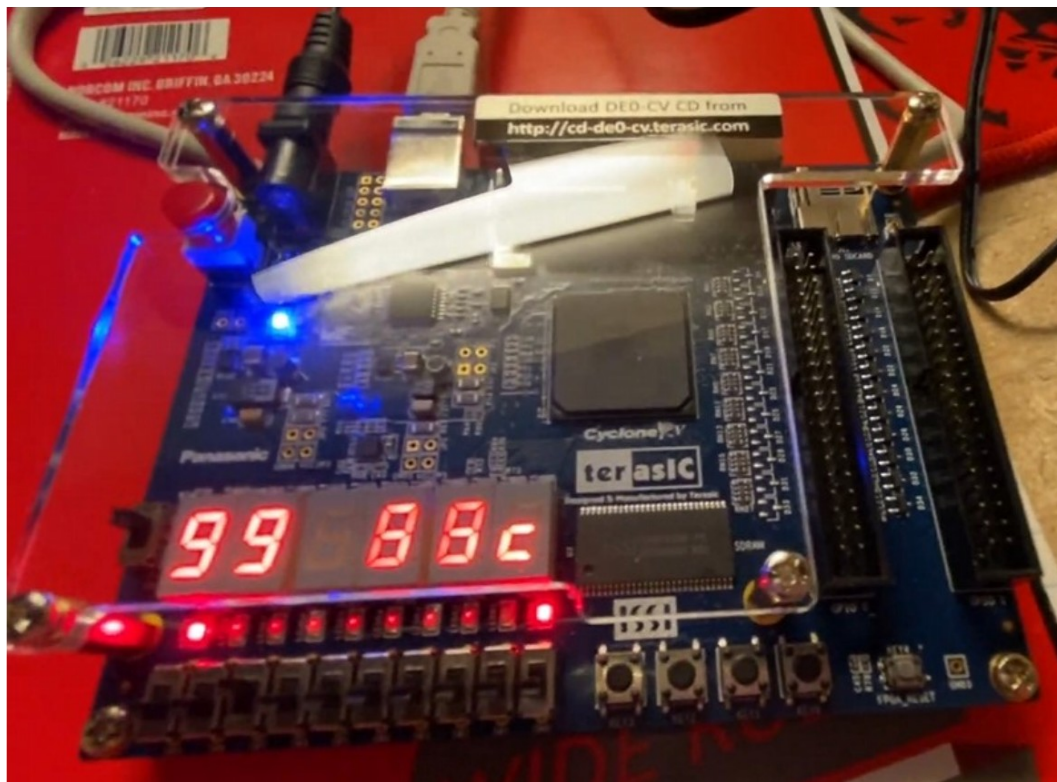


Figure 16: Current status is logged in after successfully entered password one more time